

# A WKB-based fixed-grid method for capturing trait concentration in a dispersal evolution model

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## Abstract

The evolution of dispersal traits is a fundamental topic in evolutionary ecology, where natural selection may drive the trait distribution toward concentration in the rare mutation regime. This singular behavior poses a serious numerical difficulty, since direct discretizations of the population density require very fine trait grids to identify the fittest trait and to resolve the concentrated profile accurately. In this paper, we develop a WKB-based numerical framework for a dispersal evolution model. By separating the exponentially concentrated trait dependence from a smoother amplitude variable and combining this WKB representation with dual trait-grid implementation, the method recovers the selected trait and the associated concentration structure accurately and efficiently on fixed trait grids. We establish a semi-discrete stationary fixed-grid asymptotic-preserving structure for the rare-mutation limit of the steady-state problem. Numerical experiments compare the proposed method with direct density discretizations and confirm its advantage in the small mutation regime.

**Key words.** dispersal evolution, rare mutation limit, trait concentration, WKB reformulation, Hamilton–Jacobi limit, asymptotic-preserving structure.

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## 1 Introduction

Adaptive dynamics studies the evolution of phenotypic traits under mutation, selection, and ecological feedback. In structured population models, natural selection is often manifested by the concentration of the population density on one or several fittest traits. In the rare-mutation regime, such concentration is usually described by a measure-valued limit in the trait variable, and the support of the limiting measure represents the selected phenotype. This viewpoint has been developed in selection-mutation models and in the Hamilton–Jacobi approach to adaptive dynamics [3, 6]. Among these problems, the evolution of dispersal is a particularly important example. In spatially heterogeneous environments, different dispersal rates may lead to different fitness advantages, and the selected strategy depends on the interaction between movement, resource distribution, and population competition. This question has been studied from several viewpoints in mathematical ecology, for example in reaction-diffusion models for slow dispersal, ideal free distributions, and spatial sorting [7, 9, 21].

In this work, we consider a representative dispersal evolution model of the form

$$\varepsilon \partial_t n_\varepsilon - D(\theta) \Delta_{\mathbf{x}} n_\varepsilon - \varepsilon^2 \Delta_\theta n_\varepsilon = n_\varepsilon (K(\mathbf{x}) - \rho_\varepsilon(\mathbf{x})), \quad \mathbf{x} \in \Omega. \quad (1.1)$$

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Here  $t$  denotes time,  $\mathbf{x}$  is the spatial variable, and  $\theta$  is the phenotypic trait. The unknown  $n_\varepsilon(t, \mathbf{x}, \theta)$  is the structured population density, and

$$\rho_\varepsilon(t, \mathbf{x}) = \int_{\mathbb{T}} n_\varepsilon(t, \mathbf{x}, \theta) d\theta, \quad P_\varepsilon(t, \theta) = \int_{\Omega} n_\varepsilon(t, \mathbf{x}, \theta) d\mathbf{x},$$

denote the population density in space and the trait marginal, respectively. The concentration of  $P_\varepsilon$  in the rare-mutation regime is used to identify the selected trait. The function  $K(\mathbf{x})$  represents the local carrying capacity, while the trait-dependent diffusivity  $D(\theta)$  models the dispersal rate associated with phenotype  $\theta$ . Throughout the paper, we assume that the dispersal rate is positive and periodic in the trait variable and that the carrying capacity is nonnegative and bounded:

$$0 < D_{\min} \leq D(\theta) \leq D_{\max} < \infty, \quad 0 \leq K(\mathbf{x}) \leq K_{\max}. \quad (1.2)$$

We impose homogeneous Neumann boundary conditions on  $\partial\Omega$  and periodic boundary conditions in the trait direction. The time-dependent model (1.1) is used here as an evolutionary relaxation toward the selection state, while the asymptotic structure of interest is motivated by the rare-mutation limit of the associated steady-state problem.

The steady-state version of this dispersal evolution model was studied by B. Perthame and P. E. Souganidis in [20]. Their work provides a rare-mutation description of the selected dispersal trait for a population structured by space and by a continuous dispersal trait. In the limit of vanishing mutations, the population concentrates in the trait variable, while the spatial profile remains nontrivial. This result gives the asymptotic background for the present paper and clarifies the limiting selection structure that the numerical method aims to capture.

This dispersal model fits into a broader class of structured selection models with concentration in the small-mutation limit. For purely trait-structured or nonlocal selection models, Dirac concentration and constrained Hamilton–Jacobi equations have been studied in [5] and [16]. When space is included as an additional structuring variable, the analysis becomes more delicate. Related Hamilton–Jacobi limits for populations structured by space and trait were obtained in [4], and an asymptotic analysis of a selection model with space was given in [18]. The Perthame–Souganidis dispersal model has also motivated later studies on dispersal evolution, conditional dispersal, and the stability of Dirac concentrations [8, 14, 15].

The same concentration mechanism creates a severe numerical difficulty. Direct discretizations of the original density must resolve a trait profile whose width shrinks with the mutation scale. Hence the mesh required in the trait direction may become prohibitively fine when  $\varepsilon$  is small. This suggests that the numerical method should exploit the asymptotic separation between the singular trait dependence and the smoother amplitude, rather than resolve the concentrated density directly.

This is closely related to the idea of asymptotic-preserving schemes, whose goal is to capture the limiting regime on discretizations that do not resolve the small scale  $\varepsilon$  [12]. In rare-mutation problems, the natural asymptotic separation is provided by a WKB or Hopf–Cole transformation. The density is represented by an exponential phase, which describes the concentration in the trait variable, multiplied by a smoother amplitude. Such a WKB viewpoint is standard in the Hamilton–Jacobi approach to adaptive dynamics [3] and is also central in the dispersal model of [20].

WKB-type transformations have been used as numerical tools in different multiscale regimes. For highly oscillatory problems, such as semiclassical Schrödinger equations, the exponential phase is typically imaginary or complex-valued, and the WKB transformation separates the rapid oscillations from smoother amplitude variables [2, 13]. In selection-mutation problems, by contrast, the phase is real-valued and the small parameter leads to concentration rather than oscillation. In adaptive dynamics, a WKB representation has been used to construct an asymptotic-preserving

scheme for capturing Dirac concentrations on coarse,  $\varepsilon$ -independent phenotype meshes [1]. Related Hopf–Cole based ideas have been developed for kinetic reaction–transport equations in singular front propagation regimes [10], and for kinetic BGK models in the hyperbolic limit [17]. These works motivate the WKB-based formulation adopted here for the dispersal evolution model.

The present paper develops a WKB-based fixed-grid numerical framework for the dispersal evolution problem (1.1). Starting from the phase-amplitude representation, we derive a semi-discrete scheme for the phase and amplitude variables and prove the positivity of the amplitude. We then establish a stationary fixed-grid asymptotic preserving structure associated with the rare-mutation limit of the steady-state problem under a one-sided remainder stability condition. For computation, we propose a dual trait-grid implementation. The one-dimensional phase is resolved on a fine trait grid, whereas the space-dependent amplitude is evolved on a coarser trait grid. This separation preserves the main advantage of the WKB formulation and reduces the dominant cost associated with the space-dependent unknown.

The paper is organized as follows. Section 2 recalls the WKB ansatz and the steady-state rare-mutation structure that motivates the numerical formulation. Section 3 presents the semi-discrete WKB scheme, proves positivity of the amplitude, and establishes the stationary fixed-grid asymptotic preserving structure. Section 4 describes the fully discrete implementation used in the computations and presents numerical experiments comparing the WKB method with direct density discretizations in the small-mutation regime. The appendices collect the auxiliary numerical Hamiltonians, fluxes, and remainder estimates used in the analysis. Finally, conclusions are drawn in Section 5.

## 2 Preliminary

### 2.1 WKB ansatz and reformulated system

To capture the concentration phenomenon in the trait variable, we introduce the classical WKB ansatz

$$n_\varepsilon(x, \theta, t) = W_\varepsilon(x, \theta, t) \exp\left(\frac{u_\varepsilon(\theta, t)}{\varepsilon}\right). \quad (2.3)$$

This decomposition separates the exponentially varying part in the trait direction from a comparatively smoother amplitude. It is therefore natural for both the continuous asymptotic analysis and the numerical design in the small-mutation regime.

Substituting (2.3) into (1.1) and assigning the leading trait-direction exponential contributions to the phase function, we obtain the reformulated system

$$\begin{cases} \partial_t u_\varepsilon - |\partial_\theta u_\varepsilon|^2 - \varepsilon \partial_{\theta\theta} u_\varepsilon = -\mathcal{H}(\theta, t), \\ \varepsilon \partial_t W_\varepsilon - D(\theta) \Delta_x W_\varepsilon - \varepsilon^2 \partial_{\theta\theta} W_\varepsilon - 2\varepsilon \partial_\theta W_\varepsilon \partial_\theta u_\varepsilon = W_\varepsilon (K(x) - \rho_\varepsilon(x, t) + \mathcal{H}(\theta, t)), \end{cases} \quad (2.4)$$

where  $\mathcal{H}(\theta, t)$  denotes an effective Hamiltonian. Using the identity

$$\partial_\theta (W_\varepsilon \partial_\theta u_\varepsilon) = \partial_\theta W_\varepsilon \partial_\theta u_\varepsilon + W_\varepsilon \partial_{\theta\theta} u_\varepsilon,$$

we rewrite the term  $-2\varepsilon \partial_\theta W_\varepsilon \partial_\theta u_\varepsilon$  as  $2\varepsilon \partial_\theta F_\varepsilon + 2\varepsilon W_\varepsilon \partial_{\theta\theta} u_\varepsilon$ . Consequently, the system can be written equivalently as

$$\begin{cases} \partial_t u_\varepsilon - |\partial_\theta u_\varepsilon|^2 - \varepsilon \partial_{\theta\theta} u_\varepsilon = -\mathcal{H}(\theta, t), \\ \varepsilon \partial_t W_\varepsilon - D(\theta) \Delta_x W_\varepsilon - \varepsilon^2 \partial_{\theta\theta} W_\varepsilon + 2\varepsilon \partial_\theta F_\varepsilon = W_\varepsilon (K(x) - \rho_\varepsilon(x, t) + \mathcal{H}(\theta, t) - 2\varepsilon \partial_{\theta\theta} u_\varepsilon). \end{cases} \quad (2.5)$$

In the continuous theory,  $\mathcal{H}$  is related to a principal eigenvalue problem in the spatial variable.

## 2.2 Continuous asymptotic structure of the steady state

We next recall the steady-state asymptotic structure that motivates the numerical method. Let  $(\bar{n}_\varepsilon, \bar{\rho}_\varepsilon)$  be a positive steady state of (1.1). Then

$$-D(\theta)\Delta_x \bar{n}_\varepsilon - \varepsilon^2 \partial_{\theta\theta} \bar{n}_\varepsilon = \bar{n}_\varepsilon (K(x) - \bar{\rho}_\varepsilon(x)), \quad \bar{\rho}_\varepsilon(x) = \int_{\mathbb{T}} \bar{n}_\varepsilon(x, \theta) d\theta. \quad (2.6)$$

As shown in [20], under standard assumptions on  $K$ ,  $D$ , and the domain, the steady-state family admits several properties that are fundamental for the rare-mutation limit.

**Theorem 2.1** (Continuous asymptotic structure of the steady state). *Assume that  $K$  is positive and nonconstant, that  $D$  is positive, periodic, and has a unique minimizer  $\theta_m$ , and that the steady problem (2.6) admits a positive solution. Then the following properties hold at the continuous level.*

1. *There exists a constant  $C > 0$ , independent of  $\varepsilon$ , such that*

$$0 \leq \bar{\rho}_\varepsilon(x) \leq C \quad \text{for all } x \in \Omega. \quad (2.7)$$

*Moreover, up to subsequences,  $\bar{\rho}_\varepsilon$  converges uniformly to a limit density  $\rho$ .*

2. *If we write*

$$\bar{n}_\varepsilon(x, \theta) = \exp\left(\frac{\bar{u}_\varepsilon(x, \theta)}{\varepsilon}\right), \quad (2.8)$$

*then  $\bar{u}_\varepsilon$  is uniformly Lipschitz continuous in  $\theta$ , and along subsequences*

$$\bar{u}_\varepsilon \rightarrow u \quad \text{uniformly in } (x, \theta), \quad (2.9)$$

*where the limit  $u$  is independent of  $x$ , periodic in  $\theta$ , and satisfies*

$$\begin{cases} |\partial_\theta u(\theta)|^2 = \mathcal{H}(\theta, \rho(\cdot)), \\ \max_{\theta \in \mathbb{T}} u(\theta) = 0. \end{cases} \quad (2.10)$$

3. *The effective Hamiltonian  $\mathcal{H}(\theta, \rho)$  is defined through the principal eigenvalue problem*

$$\begin{cases} -D(\theta)\Delta_x \mathcal{N}(x, \theta) = \mathcal{N}(x, \theta)(K(x) - \rho(x)) + \mathcal{N}(x, \theta) \mathcal{H}(\theta, \rho(\cdot)), & x \in \Omega, \\ \partial_\nu \mathcal{N} = 0, & x \in \partial\Omega, \end{cases} \quad (2.11)$$

*with a positive eigenfunction  $\mathcal{N}$ . Moreover,  $\mathcal{H}$  has the same monotonicity in  $\theta$  as  $D$ , and hence*

$$\min_{\theta \in \mathbb{T}} \mathcal{H}(\theta, \rho(\cdot)) = \mathcal{H}(\theta_m, \rho(\cdot)) = 0. \quad (2.12)$$

4. *The steady state concentrates on the fittest trait in the limit  $\varepsilon \rightarrow 0$ . More precisely,*

$$\bar{n}_\varepsilon \rightharpoonup N_m(x) \delta(\theta - \theta_m), \quad \bar{\rho}_\varepsilon \rightarrow N_m(x), \quad (2.13)$$

*where  $N_m$  solves the reduced Fisher-type problem*

$$\begin{cases} -D(\theta_m)\Delta_x N_m = N_m(K(x) - N_m), & x \in \Omega, \\ \partial_\nu N_m = 0, & x \in \partial\Omega. \end{cases} \quad (2.14)$$

Theorem 2.1 recalls the continuous rare-mutation structure that motivates the numerical construction. It shows that the selected trait is encoded by the limiting phase, while the spatial profile is determined through an effective Hamiltonian. The discrete counterpart of this structure will be discussed later in Section .

### 3 WKB reformulated numerical scheme

In this section, we develop and analyze a semi-discrete numerical scheme for the WKB reformulation (2.5). The scheme is formulated in terms of the phase variable  $u_\varepsilon$  and the amplitude variable  $W_\varepsilon$ , with the total density  $n_\varepsilon$  reconstructed from the WKB representation.

#### 3.1 Semi-discrete numerical discretization

We work at the semi-discrete level in order to separate the discretization in  $(x, \theta)$  from the additional issue of time stepping. For clarity, we present the scheme in one spatial dimension with uniform grids in the  $x$ - and  $\theta$ -directions. This choice is not essential for the WKB decomposition, but it keeps the notation transparent.

Let  $\{x_j\}_{j=1}^J$  be a uniform grid on  $\Omega = (a, b)$  and let  $\{\theta_k\}_{k=1}^K$  be a uniform periodic grid on  $\mathbb{T} = (0, 1)$ , with mesh size  $\Delta\theta$ . We denote

$$W_{j,k}^\varepsilon(t) \approx W_\varepsilon(t, x_j, \theta_k), \quad u_k^\varepsilon(t) \approx u_\varepsilon(t, \theta_k).$$

For notational convenience, we write

$$D_k = D(\theta_k), \quad K_j = K(x_j).$$

By (1.2), we have

$$0 < D_{\min} \leq D_k \leq D_{\max} < \infty, \quad 0 \leq K_j \leq K_{\max}.$$

For a grid function  $V = \{V_{j,k}\}$ , we use the standard second-order difference operators

$$\delta_x^2 V_{j,k} = \frac{V_{j+1,k} - 2V_{j,k} + V_{j-1,k}}{(\Delta x)^2}, \quad \delta_\theta^2 V_{j,k} = \frac{V_{j,k+1} - 2V_{j,k} + V_{j,k-1}}{(\Delta\theta)^2}.$$

The homogeneous Neumann boundary condition in  $x$  is imposed by the ghost values

$$V_{0,k} = V_{1,k}, \quad V_{J+1,k} = V_{J,k},$$

while all indices in the  $\theta$ -direction are understood periodically. For the phase variable, we also use

$$D_\theta^- u_k = \frac{u_k - u_{k-1}}{\Delta\theta}, \quad D_\theta^+ u_k = \frac{u_{k+1} - u_k}{\Delta\theta}.$$

The semi-discrete WKB system involves two numerical ingredients that are not contained in the standard second-order difference operators introduced above. The first one is the numerical Hamiltonian for the phase equation. The Hamilton–Jacobi term  $-|\partial_\theta u_\varepsilon|^2$  is approximated by

$$\widehat{H}(D_\theta^- u_k^\varepsilon, D_\theta^+ u_k^\varepsilon).$$

We assume that  $\widehat{H}$  is locally Lipschitz and consistent with the continuous Hamiltonian, namely

$$\widehat{H}(p, p) = -p^2.$$

We also assume that it is monotone in the sense that it is nondecreasing with respect to its first argument and nonincreasing with respect to its second argument. When  $\widehat{H}$  is differentiable, this condition reads

$$\partial_a \widehat{H}(a, b) \geq 0, \quad \partial_b \widehat{H}(a, b) \leq 0.$$

One typical example is the Godunov numerical Hamiltonians, shown in Appendix A.

The second ingredient is the conservative discretization of the transport coupling in the amplitude equation. Recall that the continuous flux is

$$F_\varepsilon = -W_\varepsilon \partial_\theta u_\varepsilon.$$

We denote by

$$\mathcal{D}_\theta \widehat{\mathcal{F}}(W^\varepsilon, u^\varepsilon)_{j,k}$$

a conservative discretization of  $\partial_\theta F_\varepsilon$  at  $(x_j, \theta_k)$ . Here  $\widehat{\mathcal{F}}$  approximates the flux  $-W_\varepsilon \partial_\theta u_\varepsilon$ . We assume that  $\mathcal{D}_\theta \widehat{\mathcal{F}}$  is consistent with

$$\partial_\theta F_\varepsilon = \partial_\theta (-W_\varepsilon \partial_\theta u_\varepsilon)$$

when evaluated on smooth functions, and that it is conservative in the periodic trait variable. For the positivity argument below, we further assume the quasi-positivity property

$$W_{j,k} = 0, \quad W_{j,\ell} \geq 0 \text{ for all } \ell \implies -\mathcal{D}_\theta \widehat{\mathcal{F}}(W, u)_{j,k} \geq 0. \quad (3.15)$$

This property is satisfied by standard monotone finite volume fluxes, such as the upwind flux in Appendix A.

With these two ingredients specified, we now define the discrete density and the semi-discrete WKB system. For notational convenience, set

$$E_k^\varepsilon(t) = \exp\left(\frac{u_k^\varepsilon(t)}{\varepsilon}\right). \quad (3.16)$$

The reconstructed nodal density is defined by

$$n_{j,k}^\varepsilon(t) = W_{j,k}^\varepsilon(t) E_k^\varepsilon(t). \quad (3.17)$$

As we shall see later, the evolution of  $u_\varepsilon$  and  $W_\varepsilon$  is coupled through the spatial marginal density. In the semi-discrete scheme, this marginal is computed by the composite quadrature in the trait direction. More precisely, we define

$$\rho_j^\varepsilon(t) := \Delta\theta \sum_{k=1}^K n_{j,k}^\varepsilon(t) = \Delta\theta \sum_{k=1}^K W_{j,k}^\varepsilon(t) E_k^\varepsilon(t), \quad j = 1, \dots, J. \quad (3.18)$$

This is the discrete counterpart of

$$\int_{\mathbb{T}} n_\varepsilon(t, x_j, \theta) d\theta = \int_{\mathbb{T}} W_\varepsilon(t, x_j, \theta) \exp\left(\frac{u_\varepsilon(t, \theta)}{\varepsilon}\right) d\theta.$$

Given

$$\boldsymbol{\rho}(t) = (\rho_1^\varepsilon(t), \dots, \rho_J^\varepsilon(t)),$$

the discrete effective Hamiltonian  $\mathcal{H}_k(\boldsymbol{\rho}(t))$  is determined by the discrete principal eigenvalue problem

$$-D_k \delta_x^2 \mathcal{N}_j^{(k)} = \mathcal{N}_j^{(k)} (K_j - \rho_{j,Q}^\varepsilon(t)) + \mathcal{N}_j^{(k)} \mathcal{H}_k(\boldsymbol{\rho}(t)), \quad j = 1, \dots, J. \quad (3.19)$$

Here  $\mathcal{N}^{(k)}$  is the corresponding positive eigenvector, equipped with the same discrete Neumann boundary condition in  $x$ . Its normalization does not affect  $\mathcal{H}_k$ . For definiteness, one may impose

$$\Delta x \sum_{j=1}^J \mathcal{N}_j^{(k)} = 1.$$

With the above notation, the semi-discrete WKB scheme reads

$$\begin{cases} \frac{d}{dt} u_k^\varepsilon + \widehat{H}(D_\theta^- u_k^\varepsilon, D_\theta^+ u_k^\varepsilon) - \varepsilon \delta_\theta^2 u_k^\varepsilon = -\mathcal{H}_k(\boldsymbol{\rho}(t)), & k = 1, \dots, K, \\ \varepsilon \frac{d}{dt} W_{j,k}^\varepsilon - D_k \delta_x^2 W_{j,k}^\varepsilon - \varepsilon^2 \delta_\theta^2 W_{j,k}^\varepsilon + 2\varepsilon \mathcal{D}_\theta \widehat{\mathcal{F}}(W^\varepsilon, u^\varepsilon)_{j,k} \\ = W_{j,k}^\varepsilon (K_j - \rho_j^\varepsilon(t) + \mathcal{H}_k(\boldsymbol{\rho}(t)) - 2\varepsilon \delta_\theta^2 u_k^\varepsilon), & j = 1, \dots, J, \quad k = 1, \dots, K. \end{cases} \quad (3.20)$$

We next show the equation satisfied by the reconstructed density  $n_{j,k}^\varepsilon$ . Since  $E_k^\varepsilon$  is independent of the spatial index  $j$ ,

$$\delta_x^2 n_{j,k}^\varepsilon = E_k^\varepsilon \delta_x^2 W_{j,k}^\varepsilon.$$

Moreover,

$$\varepsilon \frac{d}{dt} n_{j,k}^\varepsilon = E_k^\varepsilon \left( \varepsilon \frac{d}{dt} W_{j,k}^\varepsilon + W_{j,k}^\varepsilon \frac{d}{dt} u_k^\varepsilon \right). \quad (3.21)$$

Multiplying the first equation in (3.20) by  $E_k^\varepsilon W_{j,k}^\varepsilon$  and multiplying the second equation in (3.20) by  $\varepsilon E_k^\varepsilon$ , noticing (3.21), we can derive that  $n_{j,k}^\varepsilon$  satisfies

$$\varepsilon \frac{d}{dt} n_{j,k}^\varepsilon - D_k \delta_x^2 n_{j,k}^\varepsilon = (K_j - \rho_j^\varepsilon) n_{j,k}^\varepsilon + R_{j,k}^\varepsilon, \quad (3.22)$$

where the remainder has the form

$$R_{j,k}^\varepsilon = \varepsilon^2 E_k^\varepsilon \delta_\theta^2 W_{j,k}^\varepsilon - \varepsilon W_{j,k}^\varepsilon E_k^\varepsilon \delta_\theta^2 u_k^\varepsilon - W_{j,k}^\varepsilon E_k^\varepsilon \widehat{H}(D_\theta^- u_k^\varepsilon, D_\theta^+ u_k^\varepsilon) - 2\varepsilon E_k^\varepsilon \mathcal{D}_\theta \widehat{\mathcal{F}}(W^\varepsilon, u^\varepsilon)_{j,k}. \quad (3.23)$$

Thus the reconstructed density satisfies a semi-discrete analogue of the original density equation, with  $R_{j,k}^\varepsilon$  collecting the trait-direction discretization effects. When the numerical Hamiltonian and the conservative flux are evaluated on smooth exact functions, this remainder is consistent with the trait diffusion contribution  $\varepsilon^2 \partial_{\theta\theta} n^\varepsilon$ .

### 3.2 Basic structural property

The semi-discrete scheme (3.22) can be shown to be positive-preserving as long as the initial data is nonnegative. The detailed result is summarized as the following lemma.

**Lemma 3.1** (Positivity of the amplitude). *Assume that  $W_{j,k}^\varepsilon(0) \geq 0$  for all  $j, k$ . Then the semi-discrete amplitude equation in (3.20) preserves nonnegativity, namely*

$$W_{j,k}^\varepsilon(t) \geq 0 \quad \forall j, k, t \geq 0,$$

as long as the solution exists. Consequently,

$$n_{j,k}^\varepsilon(t) \geq 0 \quad \forall j, k, t \geq 0.$$

*Proof.* It is enough to verify the quasi-positivity of the right-hand side of the finite-dimensional ODE system for  $W^\varepsilon$ . From the amplitude equation in (3.20), we can write

$$\varepsilon \frac{d}{dt} W_{j,k}^\varepsilon = D_k \delta_x^2 W_{j,k}^\varepsilon + \varepsilon^2 \delta_\theta^2 W_{j,k}^\varepsilon - 2\varepsilon \mathcal{D}_\theta \widehat{\mathcal{F}}(W^\varepsilon, u^\varepsilon)_{j,k} + W_{j,k}^\varepsilon B_{j,k}(t),$$

where

$$B_{j,k}(t) = K_j - \rho_j^\varepsilon(t) + \mathcal{H}_k(\boldsymbol{\rho}(t)) - 2\varepsilon \delta_\theta^2 u_k^\varepsilon(t).$$

Let  $W^\varepsilon \geq 0$  and suppose that  $W_{j,k}^\varepsilon = 0$  for some  $(j, k)$ . Then, for an interior spatial point,

$$\delta_x^2 W_{j,k}^\varepsilon = \frac{W_{j+1,k}^\varepsilon + W_{j-1,k}^\varepsilon}{(\Delta x)^2} \geq 0, \quad \delta_\theta^2 W_{j,k}^\varepsilon = \frac{W_{j,k+1}^\varepsilon + W_{j,k-1}^\varepsilon}{(\Delta \theta)^2} \geq 0.$$

The same conclusion holds at the boundary because of the Neumann ghost values in  $x$ -direction and the periodicity in  $\theta$ -direction. The quasi-positivity assumption on the conservative flux operator implies that

$$-\mathcal{D}_\theta \hat{\mathcal{F}}(W^\varepsilon, u^\varepsilon)_{j,k} \geq 0.$$

Combining all the inequalities, noticing the fact that  $W_{j,k}^\varepsilon B_{j,k}(t) = 0$  since we assumed  $W_{j,k}^\varepsilon = 0$ , we finally have that, at this specific  $(j, k)$ , the following inequality holds,

$$\left. \frac{d}{dt} W_{j,k}^\varepsilon \right|_{W_{j,k}^\varepsilon = 0} \geq 0.$$

Thus the vector field points inward on the boundary of the nonnegative cone. The standard invariance criterion for finite-dimensional ODE systems implies the claim.  $\square$

### 3.3 Conditional stationary asymptotic preserving structure

We now turn to stationary semi-discrete states. The goal is to identify which parts of the continuous rare-mutation structure are retained on a fixed grid. By Lemma 3.1, stationary states obtained from nonnegative amplitudes satisfy

$$W_{j,k}^\varepsilon \geq 0, \quad n_{j,k}^\varepsilon = W_{j,k}^\varepsilon E_k^\varepsilon \geq 0.$$

All estimates below are for such nonnegative reconstructed densities.

**Weighted density balance.** Define

$$\eta_j^\varepsilon = \Delta \theta \sum_{k=1}^K \frac{n_{j,k}^\varepsilon}{D_k}. \quad (3.24)$$

Recalling (3.18), multiplying (3.22) by  $1/D_k$  and summing over the trait variable gives the stationary weighted balance equation

$$-\delta_x^2 \rho_j^\varepsilon = \eta_j^\varepsilon (K_j - \rho_j^\varepsilon) + \mathcal{R}_j^{\varepsilon,D}, \quad j = 1, \dots, J, \quad (3.25)$$

where

$$\mathcal{R}_j^{\varepsilon,D} = \Delta \theta \sum_{k=1}^K \frac{R_{j,k}^\varepsilon}{D_k}. \quad (3.26)$$

The term  $\mathcal{R}_j^{\varepsilon,D}$  is the discrete analogue of the continuous weighted mutation contribution

$$\varepsilon^2 \int_{\mathbb{T}} \frac{1}{D(\theta)} \partial_{\theta\theta} n^\varepsilon d\theta = \varepsilon^2 \int_{\mathbb{T}} n^\varepsilon \partial_{\theta\theta} \left( \frac{1}{D(\theta)} \right) d\theta,$$

where periodicity in the trait variable has been used. At the continuous level this term is bounded above by  $C\varepsilon^2 \rho^\varepsilon$ . For the split WKB discretization, however, the discrete operators act on the reconstructed density  $n_{j,k}^\varepsilon$  and an exact discrete chain rule is not available. We therefore isolate the needed one-sided control as the following assumption.



**Assumption 3.1** (One-sided weighted remainder stability). *For fixed spatial and trait grids, there exist constants  $C_{R,h} > 0$  and  $r_h \geq 0$ , independent of  $\varepsilon$ , such that every stationary state under consideration satisfies*

$$(\mathcal{R}_j^{\varepsilon,D})_+ \leq C_{R,h}\rho_j^\varepsilon + r_h, \quad j = 1, \dots, J. \quad (3.27)$$

This condition controls only the positive part of the weighted defect. A negative contribution is favorable in the maximum-principle estimate. The assumption should be understood as a structural stability condition on the trait discretization. For a density-level conservative discretization, this type of estimate follows from periodic discrete summation by parts in the trait variable; see Appendix B. For the WKB scheme used in the paper, the behavior of this estimate is examined numerically in Section 4. As we will see later, in the numerical tests, the weighted defect is typically nonpositive or has a small positive part, which supports the practical validity of the assumption.

**Proposition 3.1** (Conditional density bound for stationary semi-discrete states). *Let  $(u^\varepsilon, W^\varepsilon)$  be a stationary semi-discrete state with nonnegative reconstructed density. Assume that Assumption 3.1 holds. Then there exists a constant  $C_\rho > 0$ , independent of  $\varepsilon$ , such that*

$$0 \leq \rho_j^\varepsilon \leq C_\rho, \quad j = 1, \dots, J.$$

*Proof.* From the stationary weighted density balance (3.25), it is easy to get

$$-\delta_x^2 \rho_j^\varepsilon + \eta_j^\varepsilon \rho_j^\varepsilon \leq K_{\max} \eta_j^\varepsilon + C_{R,h} \rho_j^\varepsilon + r_h.$$

by using  $K_j \leq K_{\max}$  and Assumption 3.1. Let  $j_*$  be a point where

$$R^\varepsilon := \max_j \rho_j^\varepsilon = \rho_{j_*}^\varepsilon.$$

Since  $\delta_x^2 \rho_{j_*}^\varepsilon \leq 0$ , we have

$$\eta_{j_*}^\varepsilon \rho_{j_*}^\varepsilon \leq K_{\max} \eta_{j_*}^\varepsilon + C_{R,h} \rho_{j_*}^\varepsilon + r_h. \quad (3.28)$$

On another hand, recalling (1.2) and (3.24), it is easy to see that

$$\frac{1}{D_{\max}} \rho_j^\varepsilon \leq \eta_j^\varepsilon \leq \frac{1}{D_{\min}} \rho_j^\varepsilon. \quad (3.29)$$

Combining (3.28) and (3.29) gives

$$\frac{(R^\varepsilon)^2}{D_{\max}} \leq \left( \frac{K_{\max}}{D_{\min}} + C_{R,h} \right) R^\varepsilon + r_h.$$

The right-hand side is at most linear in  $R^\varepsilon$ , while the left-hand side is quadratic. Hence  $R^\varepsilon$  is bounded by a constant independent of  $\varepsilon$ . This proves the claim.  $\square$

**Boundedness of the effective Hamiltonian.** The density bound controls the coefficients in the discrete eigenvalue problem and hence the stationary phase equation.

**Proposition 3.2** (Uniform bound for the discrete effective Hamiltonian). *Assume that*

$$0 \leq \rho_j^\varepsilon \leq \bar{\rho}, \quad j = 1, \dots, J,$$

*where  $\bar{\rho}$  is independent of  $\varepsilon$ . Then the discrete effective Hamiltonian defined by (3.19) satisfies*

$$|\mathcal{H}_k(\boldsymbol{\rho}^\varepsilon)| \leq C_{\mathcal{H}}, \quad k = 1, \dots, K,$$

*where  $C_{\mathcal{H}}$  is independent of  $\varepsilon$ .*

*Proof.* The discrete eigenvalue problem is associated with the symmetric operator

$$-D_k \delta_x^2 - (K_j - \rho_j^\varepsilon).$$

Its principal eigenvalue has the Rayleigh representation

$$\mathcal{H}_k(\boldsymbol{\rho}^\varepsilon) = \min_{\varphi \neq 0} \frac{D_k \Delta x \sum_j |\delta_x^+ \varphi_j|^2 - \Delta x \sum_j (K_j - \rho_j^\varepsilon) \varphi_j^2}{\Delta x \sum_j \varphi_j^2}.$$

On one hand, since  $0 \leq K_j \leq K_{\max}$  and  $0 \leq \rho_j^\varepsilon \leq \bar{\rho}$ , we have

$$\mathcal{H}_k(\boldsymbol{\rho}^\varepsilon) \geq -K_{\max}.$$

On the other hand, testing the quotient with a constant vector gives

$$\mathcal{H}_k(\boldsymbol{\rho}^\varepsilon) \leq \bar{\rho}.$$

Thus

$$|\mathcal{H}_k(\boldsymbol{\rho}^\varepsilon)| \leq \max\{K_{\max}, \bar{\rho}\} =: C_{\mathcal{H}},$$

which proves the claim.  $\square$

For the stationary phase equation, the Godunov numerical Hamiltonian further gives a fixed-grid bound for the phase. Suppose that

$$\widehat{H}_G(D_\theta^- u_k^\varepsilon, D_\theta^+ u_k^\varepsilon) - \varepsilon \delta_\theta^2 u_k^\varepsilon = -\mathcal{H}_k(\boldsymbol{\rho}^\varepsilon), \quad (3.30)$$

where

$$\widehat{H}_G(p^-, p^+) = -((p^-)_-)^2 - ((p^+)_+)^2. \quad (3.31)$$

Summing the stationary phase equation over the periodic trait grid eliminates  $\delta_\theta^2 u^\varepsilon$ . Therefore

$$\Delta \theta \sum_{k=1}^K |D_\theta^+ u_k^\varepsilon|^2 = -\Delta \theta \sum_{k=1}^K \widehat{H}_G(D_\theta^- u_k^\varepsilon, D_\theta^+ u_k^\varepsilon) = \Delta \theta \sum_{k=1}^K \mathcal{H}_k(\boldsymbol{\rho}^\varepsilon) \leq C_{\mathcal{H}}.$$

Thus

$$\max_k |D_\theta^+ u_k^\varepsilon| \leq \frac{\sqrt{C_{\mathcal{H}}}}{\sqrt{\Delta \theta}}. \quad (3.32)$$

**Formal asymptotic preserving trait-selection limit.** We now pass formally to the fixed-grid rare-mutation limit. The density bound gives compactness of the discrete spatial marginal, and the Godunov phase estimate gives compactness of the normalized phase. Passing to the limit in the stationary phase equation yields a discrete constrained Hamilton–Jacobi structure.

**Proposition 3.3** (Formal asymptotic preserving structure with Godunov Hamiltonian). *Assume that the stationary semi-discrete states satisfy Assumption 3.1, and normalize the phase by*

$$\max_k u_k^\varepsilon = 0.$$

*Assume also that the stationary phase equation is discretized with  $\widehat{H}_G$  defined in (3.31). Then, up to a subsequence on the fixed spatial and trait grids,*

$$\boldsymbol{\rho}^\varepsilon \rightarrow \boldsymbol{\rho}^0, \quad \mathbf{u}^\varepsilon \rightarrow \mathbf{u}^0.$$

Moreover, the limiting phase satisfies the discrete constrained Hamilton–Jacobi system

$$\widehat{H}_G(D_\theta^- u_k^0, D_\theta^+ u_k^0) = -\mathcal{H}_k(\boldsymbol{\rho}^0), \quad \max_k u_k^0 = 0.$$

If further  $\mathcal{H}_k(\boldsymbol{\rho}^0)$  has a unique zero at  $k_m$  on the fixed trait grid, then

$$k^* = k_m,$$

where  $k^* = \arg \max_k u_k^0$ , meaning that the selected grid trait is exactly  $\theta_{k_m}$ .

*Proof.* Proposition 3.1 gives a uniform bound on  $\boldsymbol{\rho}^\varepsilon$ . Since the spatial grid is fixed, a subsequence converges to some  $\boldsymbol{\rho}^0$ . The preceding Godunov phase estimate and the normalization give a uniform bound on  $\mathbf{u}^\varepsilon$  on the fixed trait grid. Hence, up to a further subsequence,

$$\mathbf{u}^\varepsilon \rightarrow \mathbf{u}^0.$$

Passing to the limit in the stationary phase equation gives the limiting discrete constrained Hamilton–Jacobi system, because  $\varepsilon \delta_\theta^2 u_k^\varepsilon \rightarrow 0$  on the fixed trait grid and

$$\mathcal{H}_k(\boldsymbol{\rho}^\varepsilon) \rightarrow \mathcal{H}_k(\boldsymbol{\rho}^0).$$

The normalization also passes to the limit.

It remains to identify the selected grid trait. Let  $k_*$  be a maximizer of  $u^0$ . Then

$$D_\theta^- u_{k_*}^0 \geq 0, \quad D_\theta^+ u_{k_*}^0 \leq 0.$$

By the definition of the Godunov Hamiltonian,

$$\widehat{H}_G(D_\theta^- u_{k_*}^0, D_\theta^+ u_{k_*}^0) = 0.$$

The limiting phase equation therefore gives

$$\mathcal{H}_{k_*}(\boldsymbol{\rho}^0) = 0.$$

If  $\mathcal{H}_k(\boldsymbol{\rho}^0)$  has a unique zero at  $k_m$ , then every maximizer  $k_*$  of  $u^0$  must satisfy  $k_* = k_m$ .  $\square$

The fixed-grid asymptotic preserving structure obtained above is the discrete analogue of the continuous rare-mutation structure summarized in Theorem 2.1. The discrete density estimate in Proposition 3.1 plays the role of the continuous bound on the spatial marginal. The discrete eigenvalue problem (3.19) defines the numerical effective Hamiltonian, corresponding to the continuous principal eigenvalue problem. By Proposition 3.3, the limiting discrete Hamilton–Jacobi system identifies the selected grid trait through the maximizer of the phase, in the same way that the continuous constrained Hamilton–Jacobi equation identifies the selected trait in the rare-mutation limit. Thus the semi-discrete WKB scheme preserves the essential selection structure on fixed spatial and trait grids.

The preceding argument is a stationary fixed-grid asymptotic preserving structure. In the computations, the time-dependent WKB system is used as a relaxation dynamics toward such stationary states. Thus the semi-discrete analysis identifies the limiting stationary structure, while the numerical algorithm evolves the WKB variables toward this discrete eigenvalue and constrained Hamilton–Jacobi structure.

## 4 Numerical experiments

This section presents the fully discrete implementation and numerical experiments for the WKB formulation. We focus on its behavior in the small-mutation regime and compare it with direct density discretizations to demonstrate the advantage of the WKB formulation.

### 4.1 Fully discrete implementation and dual trait-grid implementation

We first describe the time discretization used in the numerical experiments. For comparison, the original density equation is discretized by the linear implicit update

$$\varepsilon \frac{n_{j,k}^{n+1} - n_{j,k}^n}{\tau} - D_k \delta_x^2 n_{j,k}^{n+1} - \varepsilon^2 \delta_\theta^2 n_{j,k}^{n+1} = n_{j,k}^{n+1} (K_j - \rho_j^n), \quad \rho_j^n = \Delta \theta \sum_k n_{j,k}^n. \quad (4.33)$$

For the WKB formulation, the phase variable  $u$  is advanced by

$$\frac{u_k^{n+1} - u_k^n}{\tau} + \widehat{H}(D_\theta^- u_k^n, D_\theta^+ u_k^n) - \varepsilon \delta_\theta^2 u_k^{n+1} = -H_k^n, \quad (4.34)$$

where  $H_k^n$  is determined by the discrete principal eigenvalue problem (3.19) with  $\rho_j^n$  reconstructed via (3.18). With  $u^{n+1}$  fixed, the amplitude  $W$  is advanced by the linear implicit update

$$\begin{aligned} \varepsilon \frac{W_{j,k}^{n+1} - W_{j,k}^n}{\tau} - D_k \delta_x^2 W_{j,k}^{n+1} - \varepsilon^2 \delta_\theta^2 W_{j,k}^{n+1} + 2\varepsilon D_\theta \widehat{F}(W^{n+1}, u^{n+1})_{j,k} \\ = W_{j,k}^{n+1} (K_j - \rho_j^n + H_k^n - 2\varepsilon \delta_\theta^2 u_k^{n+1}). \end{aligned} \quad (4.35)$$

Since  $u^{n+1}$  is already known, the amplitude update is also linear.

**Dual trait-grid implementation.** To improve computational efficiency, we introduce the dual trait-grid implementation. The phase variable  $u$  and the amplitude variable  $W$  are represented on different trait grids. Let  $\Theta_u$  be the fine phase grid and  $\Theta_W$  the coarser amplitude grid. When  $\Theta_u = \Theta_W$ , the method reduces to the standard single-grid WKB implementation. When  $\Theta_W$  is coarser, the amplitude  $W$  is stored and evolved on fewer trait nodes, while the phase  $u$  still resolves the selected trait on the finer one-dimensional grid.

The coupling between the two trait grids is handled by periodic interpolation in the trait variable. Since  $u$  depends only on the trait variable, refining  $\Theta_u$  is relatively inexpensive. By contrast,  $W$  depends on both space and trait. In  $d$  spatial dimensions, the number of unknowns scales as

$$O\left(N_x^d N_\theta^{(W)} + N_\theta^{(u)}\right),$$

up to the cost of the spatial solvers. When  $N_\theta^{(W)} \ll N_\theta^{(u)}$ , the dual-grid strategy reduces the dominant memory and CPU cost while retaining fine trait resolution for the selection process.

### 4.2 Numerical experiments in 1D

Unless otherwise stated, the one-dimensional tests use

$$\Omega = (0, 1), \quad \mathbb{T} = [0, 1),$$

with homogeneous Neumann boundary conditions in the spatial variable and periodic boundary conditions in the trait variable. The carrying capacity and the dispersal rate are chosen as

$$K(x) = K_0 - K_1 \cos(2\pi x), \quad D(\theta) = D_{\min} + D_1(1 - \cos(2\pi(\theta - \theta_m))), \quad (4.36)$$

respectively, with  $K_0 = 1$ ,  $K_1 = 0.5$ ,  $D_{\min} = 0.2$ ,  $D_1 = 0.4$ ,  $\theta_m = 0.35$ . Thus  $D$  has a unique minimum at  $\theta_m$ , which is the expected selected trait in the rare-mutation limit. The default initial data are chosen as

$$u_k^0 = 0, \quad W_{j,k}^0 = n_{j,k}^0 = 1. \quad (4.37)$$

Other initial data are specified in the individual tests when needed.

#### 4.2.1 Numerical verification of Assumption 3.1

We first examine the one-sided weighted remainder condition used in the stationary asymptotic-preserving argument. For each computation, after the solution has reached a numerical steady state, we record the pairs

$$(\rho_j^\varepsilon, \mathcal{R}_j^{\varepsilon,D}), \quad j = 1, \dots, J,$$

Figure ?? shows representative scatter plots of  $\mathcal{R}_j^{\varepsilon,D}$  against  $\rho_j^\varepsilon$  for several values of  $\varepsilon$  and for different dispersal functions  $D$ . The weighted defect is mostly nonpositive in the tested cases, and the positive part remains small relative to  $\rho_j^\varepsilon$ . This supports the practical validity of the one-sided remainder condition for the split WKB scheme used in the computations.

#### 4.2.2 Consistency between the direct density solver and the WKB formulation

We first perform a same-grid consistency check at finite  $\varepsilon$ . The direct density solver and the WKB solver are run on the same spatial and trait grids. The WKB density is reconstructed by

$$n_{j,k}^{\text{WKB}} = W_{j,k} \exp(u_k/\varepsilon),$$

and the corresponding spatial marginal and trait marginal are compared with those obtained from the direct density computation.

Figure 1 shows the comparison at  $\varepsilon = 10^{-3}$ ,  $t = 10$ ,  $J = 100$ , and  $K = 1024$ . The two solutions are almost indistinguishable for both the spatial marginal  $\rho(x)$  and the trait marginal  $P(\theta)$ . This indicates that, when the trait grid is sufficiently fine, the WKB formulation is consistent with the direct density formulation at the density level.

Table 1 further reports the same-grid errors as the trait grid is refined. The errors decrease steadily with  $K$ . After the trait grid is moderately refined, the error is reduced by roughly a factor of four whenever  $K$  is doubled, which is consistent with the second-order trait discretization. This test confirms that the WKB phase-amplitude formulation does not introduce a density-level inconsistency when the direct density profile is sufficiently resolved.

Table 1: Same-grid consistency between the direct density solver and the WKB reconstruction at  $\varepsilon = 10^{-3}$ ,  $t = 10$ , and  $J = 100$ .

| $K$        | 16      | 32      | 64      | 128     | 256     | 512     | 1024    | 2048    |
|------------|---------|---------|---------|---------|---------|---------|---------|---------|
| $L^2$      | 1.74E-1 | 3.84E-2 | 8.70E-3 | 2.13E-3 | 5.30E-4 | 1.32E-4 | 3.30E-5 | 8.21E-6 |
| $L^\infty$ | 1.42E-1 | 4.59E-2 | 1.03E-2 | 2.63E-3 | 6.54E-4 | 1.63E-4 | 4.08E-5 | 1.02E-5 |

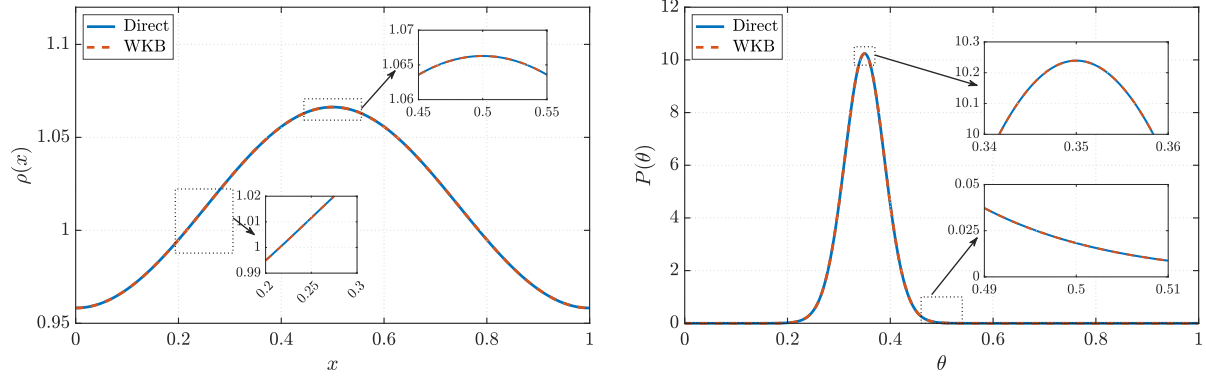


Figure 1: Same-grid comparison between the direct density solver and the WKB formulation. Left: spatial marginal  $\rho(x)$ . Right: trait marginal  $P(\theta)$ . Parameters are  $\varepsilon = 10^{-3}$ ,  $t = 10$ ,  $J = 100$ , and  $K = 1024$ .

#### 4.2.3 Long-time concentration in the trait direction

Here we illustrate the concentration process in the trait direction. According to the rare-mutation asymptotics, the trait marginal  $P(\theta, t)$  should become increasingly concentrated near the selected trait as time evolves, and the concentration should become sharper when  $\varepsilon$  decreases.

Figure 2 shows this behavior. In the left panel,  $\varepsilon$  is fixed and the trait marginal is plotted at several time levels. The peak gradually moves toward the expected selected trait  $\theta_m = 0.35$  and becomes sharper as time increases (Here we should say that it approaches a steady profile, which might be sharp). In the right panel, the final-time marginals are compared for different values of  $\varepsilon$ . The concentration becomes more pronounced as  $\varepsilon$  decreases. These results are consistent with the continuous rare-mutation structure and show that the WKB formulation captures the emergence of the selected trait.

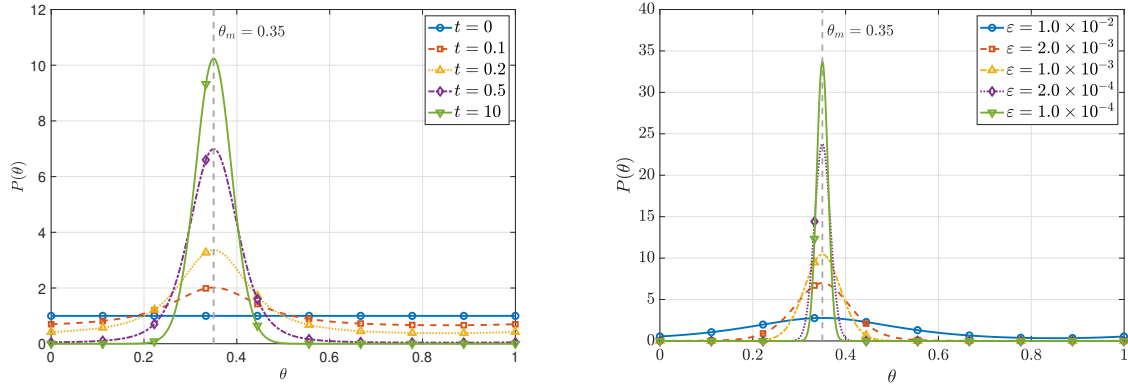


Figure 2: Long-time concentration in the trait direction. Left: evolution of  $P(\theta, t)$  for fixed  $\varepsilon = 10^{-3}$ ,  $K = 128$ , and  $\tau = 10^{-3}$ . Right: final-time trait marginals for different values of  $\varepsilon$ . The vertical dashed line marks the expected selected trait  $\theta_m$ .

#### 4.2.4 WKB versus direct density discretization on coarse trait grids

We next compare the WKB formulation with direct density discretizations on coarse trait grids, aiming to illustrate the main numerical advantage of the WKB representation. In the small-mutation regime, the direct density becomes sharply concentrated in the trait direction, and a coarse trait grid may fail to resolve the location and shape of the concentration. By contrast, in the WKB formulation the selected trait is encoded in the phase variable, which remains much smoother than the density.

Figure 3 shows the comparison for decreasing values of  $\varepsilon$ . The direct density solutions are plotted for several trait resolutions, while the WKB result is shown as the reference WKB computation on the same physical problem. For moderate  $\varepsilon$ , both approaches give comparable results once the trait grid is sufficiently refined. As  $\varepsilon$  decreases, however, the direct density solver becomes much more sensitive to the trait resolution. In particular, coarse direct grids give visibly shifted or broadened trait marginals. The WKB formulation remains well aligned with the selected trait because the phase variable resolves the concentration location without directly resolving the full density profile.

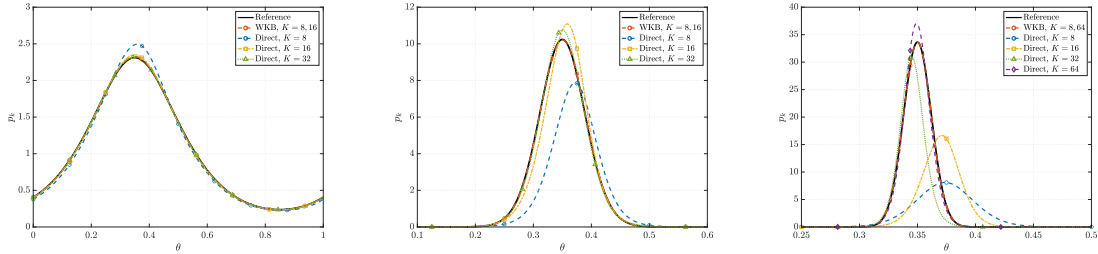


Figure 3: Comparison between the WKB formulation and direct density discretizations on coarse trait grids. From left to right, the mutation parameter decreases. The direct density solver requires increasingly fine trait grids to resolve the concentrated marginal, whereas the WKB formulation continues to capture the selected trait through the phase variable. (We might use post-processing for  $\varepsilon = 64$  since the current choice seems that  $K$  is still large)

This experiment shows that the benefit of the WKB formulation becomes more pronounced as  $\varepsilon$  decreases. The result is consistent with the asymptotic picture: the density-level trait marginal develops a narrow concentration, while the WKB phase remains a more regular quantity from which the selected trait can be identified on a fixed trait grid.

To quantify the concentration profile, we further extract several peak diagnostics from the trait marginal  $P(\theta)$ . The same postprocessing is used for all data sets. Since  $P(\theta)$  is sharply concentrated when  $\varepsilon$  is small, the interpolation is performed on  $\log P$  rather than on  $P$  itself. We use periodic spline interpolation in the trait variable and locate the maximum of the interpolated  $\log P$ . From this local reconstruction we compute the peak location, the peak height, the full width at half maximum, and a local logarithmic-curvature indicator.

Table 2 reports the errors of the WKB peak diagnostics with respect to a refined WKB reference computation. The errors decrease systematically as the trait grid is refined. In particular, the error in the peak location is already below  $5 \times 10^{-4}$  for  $K_u = K_W = 16$  and decreases to the order of  $10^{-9}$  by  $K_u = K_W = 512$ . The errors in the peak height and width also decrease steadily. This shows that the WKB formulation gives stable peak information even on relatively coarse trait grids.

Table 3 gives the corresponding diagnostics for the direct density solver, using a refined WKB computation as the benchmark. The direct method improves as the trait grid is refined, but the convergence is less regular. The peak location error decreases only when the trait grid becomes

sufficiently fine, and the height and curvature-based quantities are more sensitive to the grid. This behavior is expected, since the direct density solver has to resolve the concentrated trait marginal itself. The comparison confirms that the WKB phase-amplitude representation is more effective for extracting the selected trait and the local concentration structure in the small-mutation regime.

Table 2: Errors of the WKB peak diagnostics relative to the refined WKB reference computation with  $K_u = K_W = 2048$ . The reference values are  $\theta_*^{\text{ref}} = 0.350$ ,  $P_{\text{max}}^{\text{ref}} = 10.5$ ,  $w_{\text{FWHM}}^{\text{ref}} = 9.20\text{E-}2$ ,  $\kappa_{\text{log}}^{\text{ref}} = 6.69\text{E}2$ , and  $w_{\text{curv}}^{\text{ref}} = 9.11\text{E-}2$ . The comparison is made at  $t = 10$ . (I suggest we add a column for the CPU time)

| $K_u$ | $K_W$ | $e_\theta$ | $e_P$   | $e_{\text{FWHM}}$ | $e_\kappa$ | $e_{w,\text{curv}}$ |
|-------|-------|------------|---------|-------------------|------------|---------------------|
| 16    | 16    | 4.84E-4    | 3.86E-3 | 4.54E-3           | 2.43E-2    | 1.24E-2             |
| 32    | 32    | 4.05E-5    | 8.58E-4 | 1.05E-3           | 3.33E-3    | 1.66E-3             |
| 64    | 64    | 1.99E-6    | 2.03E-4 | 2.22E-4           | 6.42E-4    | 3.21E-4             |
| 128   | 128   | 4.83E-7    | 5.38E-5 | 5.67E-5           | 1.51E-4    | 7.56E-5             |
| 256   | 256   | 3.04E-8    | 1.31E-5 | 1.37E-5           | 3.64E-5    | 1.82E-5             |
| 512   | 512   | 6.56E-9    | 2.68E-6 | 2.81E-6           | 1.20E-5    | 5.98E-6             |

Table 3: Errors of the direct density peak diagnostics relative to the refined WKB benchmark with  $K_u = K_W = 1024$ . The reference values are  $\theta_*^{\text{ref}} = 0.350$ ,  $P_{\text{max}}^{\text{ref}} = 10.3$ ,  $w_{\text{FWHM}}^{\text{ref}} = 9.17\text{E-}2$ ,  $\kappa_{\text{log}}^{\text{ref}} = 6.73\text{E}2$ , and  $w_{\text{curv}}^{\text{ref}} = 9.08\text{E-}2$ . The comparison is made at  $t = 10$ . (I suggest we add a column for the CPU time)

| $K_n$ | $e_\theta$ | $e_P$   | $e_{\text{FWHM}}$ | $e_\kappa$ | $e_{w,\text{curv}}$ |
|-------|------------|---------|-------------------|------------|---------------------|
| 16    | 2.50E-2    | 5.18E-2 | 8.92E-2           | 1.03E0     | 2.99E-1             |
| 32    | 6.23E-3    | 4.24E-2 | 4.70E-2           | 4.61E-1    | 1.73E-1             |
| 64    | 6.23E-3    | 2.93E-3 | 1.01E-2           | 1.25E-1    | 6.88E-2             |
| 128   | 1.59E-3    | 2.44E-3 | 3.17E-3           | 8.05E-2    | 3.80E-2             |
| 256   | 1.59E-3    | 4.35E-3 | 6.06E-4           | 2.01E-1    | 1.19E-1             |
| 512   | 3.66E-4    | 4.59E-3 | 1.61E-4           | 8.15E-2    | 3.84E-2             |
| 1024  | 3.66E-4    | 4.71E-3 | 4.67E-6           | 2.06E-1    | 1.22E-1             |

#### 4.2.5 Spatial and temporal accuracy tests

(Here we aim to show the spatial and temporal accuracy of WKB. )

#### 4.3 Two-dimensional spatial experiment

(A 2D example is presented to show the accuracy and efficiency of our method.)

## 5 Conclusion

### A Numerical Hamiltonians and conservative fluxes

For the monotone theory in Section 3, the numerical Hamiltonian  $\hat{H}(a, b)$  is assumed to be consistent with  $-p^2$ , nondecreasing in its first argument, and nonincreasing in its second argument. One typical choice is the Godunov Hamiltonian

$$\hat{H}_G(a, b) = -(a^-)^2 - (b^+)^2, \quad a^- := \min\{a, 0\}, \quad b^+ := \max\{b, 0\}. \quad (1.38)$$



In practice, high-order one-sided WENO reconstructions can be applied [11, 19, 22]. This is a numerical option for improving resolution of the phase profile.

The conservative flux in the equation of  $W$  can be chosen as

$$D_\theta \widehat{F}(W, u)_{j,k} = \frac{\widehat{F}_{j,k+1/2} - \widehat{F}_{j,k-1/2}}{\Delta\theta}, \quad a_{k+1/2} = -\frac{u_{k+1} - u_k}{\Delta\theta},$$

with the upwind flux

$$\widehat{F}_{j,k+1/2}^{\text{up}} = a_{k+1/2}^+ W_{j,k} + a_{k+1/2}^- W_{j,k+1}, \quad a^+ = \max\{a, 0\}, \quad a^- = \min\{a, 0\}. \quad (1.39)$$

## B A sufficient condition for the one-sided remainder bound

We give a simple situation in which Assumption 3.1 can be verified. Suppose that the reconstructed density satisfies the conservative semi-discrete density equation

$$\varepsilon \frac{d}{dt} n_{j,k}^\varepsilon - D_k \delta_x^2 n_{j,k}^\varepsilon - \varepsilon^2 \delta_\theta^2 n_{j,k}^\varepsilon = (K_j - \rho_{j,Q}^\varepsilon) n_{j,k}^\varepsilon. \quad (2.40)$$

Then periodic summation by parts gives

$$\mathcal{R}_j^{\varepsilon,D} = \varepsilon^2 \Delta\theta \sum_k n_{j,k}^\varepsilon \delta_\theta^2 \left( \frac{1}{D_k} \right).$$

In this case,

$$(R_j^{\varepsilon,D})_+ \leq \varepsilon^2 C_{D,h} \rho_j^\varepsilon, \quad C_{D,h} = \max_k \left| \delta_\theta^2 \left( \frac{1}{D_k} \right) \right|.$$

This verifies Assumption 3.1 with  $C_{R,h} = C_{D,h}$  for  $\varepsilon \in (0, 1]$  and  $r_h = 0$ .

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